

Qatar-1b

R-0409

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_170921 · created 2026-07-19T05:39:58+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458017.6542 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.127 +/- 0.054
Transit depth (Rp/Rs)^2	1.61 +/- 1.38 [%]
Orbital Inclination (inc)	80.26 +/- 2.93
Airmass coefficient 1 (a1)	70.22 +/- 44.56
Airmass coefficient 2 (a2)	-3.0 +/- 1.42
Scatter in the residuals of the lightcurve fit is	60.74 %
Best Comparison Star	#1 - [539, 126]
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.029 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.127 $\pm$ 0.054 vs 0.1463 (deviation 0.32 σ)
Duration fit vs published	0.029 d vs 0.0692 d (deviation 1.27 σ)
Mid-time O-C	0.5 min
Depth SNR	2.1
Residual scatter	60.74 %

Light curve

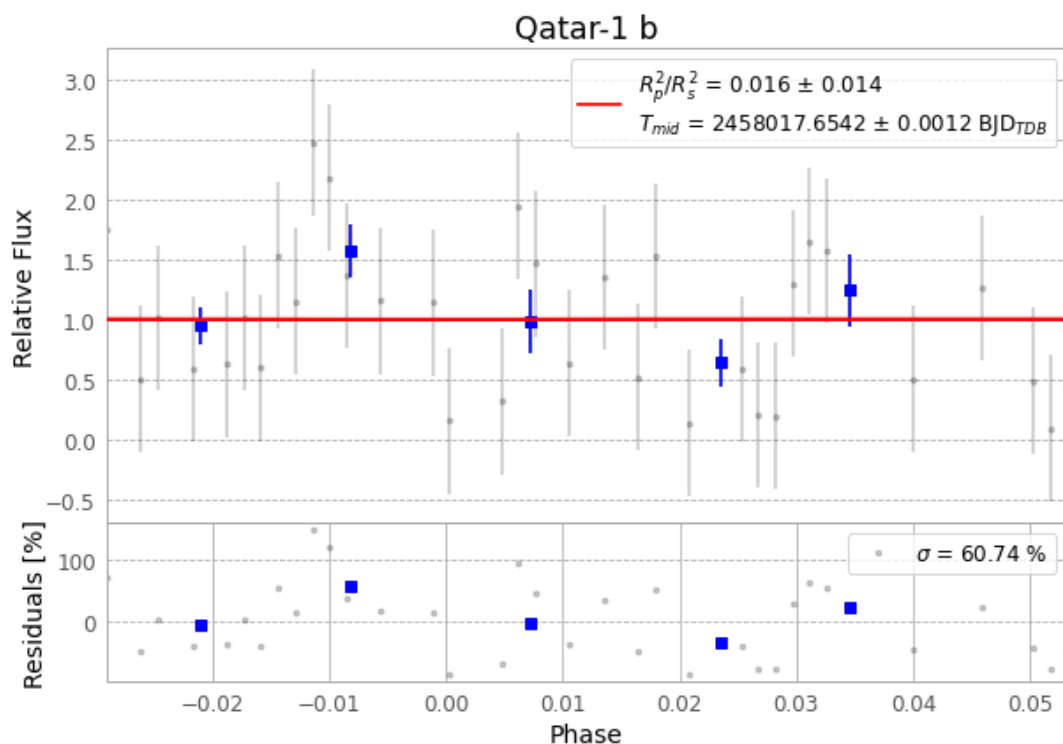


Figure 1 — FinalLightCurve_Qatar-1 b_2017-09-20.png (SHA-256 79d59c51542dc65e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 434], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 71477b6e9948ce06...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

58 FITS frames (38 MB), Qatar-1170921023912.FITS → Qatar-1170921053012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-170921.log (4a11ea76da3f...), FinalLightCurve_Qatar-1 b_2017-09-20.png (79d59c51542d...), FinalLightCurve_Qatar-1 b_2017-09-20.csv (c43bf8be01ac...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 05:39Z.